

DATA MINING

Section 1: MCQ (Multiple Choice Questions)

1. Choose which method is commonly used for data gathering in the field of Human-Computer Interaction?

- A. Observing users in their natural environment.
- B. Conducting surveys and questionnaires.
- C. Analyzing computer code.
- D. All of the these.

Answer: Observing users in their natural environment.

2. Discover the primary goal of interaction design.

- A. To create aesthetically pleasing interfaces.
- B. To design interfaces that are easy and intuitive for users to interact with.
- C. To develop complex algorithms for computer interaction.
- D. To increase the processing speed of computers.

Answer: To design interfaces that are easy and intuitive for users to interact with.

3. Locate what does the term \"dimensional modeling\" refer to in data warehousing?

- A. Modeling 3D graphics for visualization
- B. Designing databases for online transactions
- C. Organizing data for analytical purposes
- D. Processing real-time data streams

Answer: Organizing data for analytical purposes

4. Select which of the following is NOT a component of a data warehouse architecture?

- A. Data visualization tools
- B. ETL processes
- C. Data warehouse server
- D. Metadata repository

Answer: Data visualization tools

5. Identify the purpose of a data mart in data warehousing?

- A. Storing raw data
- B. Providing a subset of data for a specific business unit
- C. Processing online transactions
- D. Generating real-time reports

Answer: Providing a subset of data for a specific business unit

6. Select which term refers to the process of transforming data into a common format for storage in a data warehouse?

- A. Data extraction
- B. Data transformation
- C. Data loading
- D. Data modeling

Answer: Data transformation

7. Identify which of the following is a characteristic of data warehouses compared to operational databases?

- A. Optimized for transaction processing
- B. Subject-oriented
- C. Supports day-to-day operations
- D. Contains current data only

Answer: Subject-oriented

8. Select what is the primary source of data for a data warehouse?

- A. Online transaction processing (OLTP) systems
- B. Social media platforms
- C. Real-time data streams
- D. External data sources

Answer: Online transaction processing (OLTP) systems

9. Select the purpose of a metadata repository in data warehousing?

- A. Storing historical data
- B. Managing data access permissions
- C. Describing and organizing warehouse data
- D. Generating real-time reports

Answer: Describing and organizing warehouse data

10. Define the term "granularity" refer to in data warehousing?

- A. The size of a data warehouse server
- B. The level of detail in the data
- C. The speed of data retrieval
- D. The frequency of data updates

Answer: The level of detail in the data

11. Identify the purpose of data replication in data warehousing?

- A. Reducing data redundancy
- B. Increasing data availability
- C. Enhancing data security
- D. Storing historical data

Answer: Increasing data availability

12. Select what role does data cleansing play in data warehousing?

- A. Adding noise to data
- B. Increasing data redundancy
- C. Removing inconsistencies from data
- D. Encrypting sensitive data

Answer: Removing inconsistencies from data

13. Select what is the purpose of indexing in data warehousing?

- A. Reducing data redundancy
- B. Improving data security
- C. Enhancing data retrieval performance
- D. Storing historical data

Answer: Enhancing data retrieval performance

14. Identify Which of the following is a benefit of using data warehousing for reporting and analysis?

- A. Decreased data availability
- B. Limited data accessibility
- C. Improved decision-making processes
- D. Slower data retrieval

Answer: Improved decision-making processes

15. Select of the following OLAP operations involves viewing data from different angles or perspectives?

- A. Slice
- B. Dice
- C. Drill Down
- D. Roll up

Answer: Dice

16. _____ is a subject-oriented, integrated, time-variant, nonvolatile collection of data in support of management decisions.

- A. Data Mining.
- B. Data Warehousing.

- C. Web Mining.
- D. Text Mining

Answer: Data Warehousing.

17. Exemplifying Data Warehouse is called as _____ warehouse.

- A. Read only.
- B. Write only.
- C. Read-write only.
- D. None.

Answer: Read only.

18. The important aspect of the data warehouse environment is that data found within the Data Warehouse is _____.

- A. Subject-oriented.
- B. Time-variant.
- C. Integrated.
- D. All of these.

Answer: All of these.

19. The time horizon in Datawarehouse is usually _____.

- A. 1-2 years.
- B. 3-4years.
- C. 5-6 years.
- D. 5-10 years.

Answer: 5-10 years.

20. The data is stored, retrieved & updated in _____.

- A. OLAP.
- B. OLTP.
- C. SMTP.
- D. FTP.

Answer: OLTP.

21. _____ describes the data contained in the data warehouse.

- A. A. Relational data.
- B. B. Operational data.
- C. C. Metadata.
- D. D. Informational data.

Answer: C. Metadata.

22. _____ predicts future trends & behaviors, allowing business managers to make proactive, knowledge-driven decisions.

- A. A. Data Warehouse.
- B. B. Data Mining.
- C. C. Data Marts.
- D. D. Metadata

Answer: B. Data Mining.

23. _____ is the heart of the warehouse.

- A. A. Data mining database servers.
- B. B. Data warehouse database servers.
- C. C. Data mart database servers.
- D. D. Relational data base servers.

Answer: B. Data warehouse database servers.

24. _____ is the specialized data warehouse database.

- A. A. Oracle.
- B. B. DBZ.
- C. C. Informix.
- D. D. Redbrick.

Answer: D. Redbrick.

25. _____ is held in the catalog of the warehouse database system.

- A. A. Application level metadata.
- B. B. Algorithmic level metadata.
- C. C. Departmental level metadata.
- D. D. Core warehouse metadata

Answer: B. Algorithmic level metadata.

26. The record cannot be updated in _____.

- A. A. OLTP
- B. B. Files
- C. C. RDBMS
- D. D. Data Warehouse

Answer: D. Data Warehouse

27. The source of all data warehouse data is the _____.

- A. A. Operational environment.
- B. B. Informal environment.

- C. C. Formal environment.
- D. D. Technology environment.

Answer: A. Operational environment.

28. Data warehouse contains _____ data that is never found in the operational environment.

- A. A. Normalized.
- B. B. Informational.
- C. C. Summary.
- D. D. Denormalized

Answer: C. Summary.

29. Data redundancy between the environments results in less than _____ percent.

- A. A. one.
- B. B. two.
- C. C. three.
- D. D. four.

Answer: A. one.

30. Bill Inmon has estimated _____ of the time required to build a data warehouse, is consumed in the conversion process.

- A. A. 10 percent.
- B. B. 20 percent.
- C. C. 40 percent
- D. D. 80 percent.

Answer: D. 80 percent.

31. Detail data in a single fact table is otherwise known as _____.

- A. A. Monoatomic data.
- B. B. Diatomic data.
- C. C. Atomic data.
- D. D. Multiatomic data.

Answer: C. Atomic data.

32. _____ test is used in an online transactional processing environment.

- A. A. MEGA.
- B. B. MICRO.
- C. C. MACRO.
- D. D. ACID.

Answer: D. ACID.

33. _____ is a good alternative to the star schema.

- A. A.Starschema.
- B. B.Snowflake schema.
- C. C. Fact constellation.
- D. D.Star-snowflake schema.

Answer: C. Fact constellation.

34. _____ is data that is distilled from the low level of detail found at the current detailed leve.

- A. A.Highlysummarized data.
- B. B.Lightlysummarized data.
- C. C. Metadata.
- D. D. Older detail data

Answer: B.Lightlysummarized data.

35. Attributing a data warehouse is an_____.

- A. A. Updated end users.
- B. B.Containsnumerousnamingconventions and formats
- C. C. Organized Subject area.
- D. D. Contains only current data.

Answer: C. Organized Subject area.

36. Query tool is meant for _____.

- A. A. Data acquisition.
- B. B.Informationdelivery.
- C. C.Information exchange.
- D. D.Communication.

Answer: A. Data acquisition.

37. The full form of KDD is _____.

- A. A.KnowledgeDatabase.
- B. B.KnowledgeDiscoveryinDatabase.
- C. C.KnowledgeData house.
- D. D.KnowledgeDataDefinition.

Answer: B.KnowledgeDiscoveryinDatabase.

38. The first International conference on KDD was held in the year _____.

- A. A. 1996.
- B. B.1997.

C. C.1995 .

D. D. 1994

Answer: C.1995 .

39. Removing duplicate records is a process called _____.

A. A.Recovery.

B. B. Data cleaning

C. . C.Data cleansing.

D. D.Datapruning.

Answer: B. Data cleaning

40. _____ contains information that gives users an easy-to-understand perspective of the information stored in the data warehouse.

A. A. Business metadata.

B. B.Technicalmetadata.

C. C. Operational metadata.

D. D. Financial metadata

Answer: A. Business metadata.

41. _____ helps to integrate, maintain and view the contents of the data warehousing system.

A. A.Businessdirectory.

B. B.Informationdirectory.

C. C.Datadictionary.

D. D. Database.

Answer: B.Informationdirectory.

42. . Discovery of cross-sales opportunities is called _____.

A. A.Segmentation.

B. B.Visualization.

C. C. Correction.

D. D.Association.

Answer: D.Association.

43. Data marts that incorporate data mining tools to extract sets of data are called _____.

A. A. Independent data mart.

B. B. Dependent data marts.

C. C. Intra-entry data mart.

D. D. Inter-entry data mart.

Answer: B. Dependent data marts.

44. _____ can generate programs itself, enabling it to carry out new tasks.

- A. A. Automated system.
- B. B. Decision-making system.
- C. C. Self learning system.
- D. D. Productivity system.

Answer: D. Productivity system.

45. . The power of self-learning system lies in _____.

- A. A. Cost.
- B. B. Speed.
- C. C. Accuracy.
- D. D. Smplicity.

Answer: C. Accuracy.

46. Critiquing which is not a component of a data warehouse?

- A. A. Metadata.
- B. B. Current detail data.
- C. C. Lightly summarized data.
- D. D. Component Key

Answer: D. Component Key

47. Calculate the components in a data warehouse.

- A. A. Two.
- B. B. Three.
- C. C. Four.
- D. D. Five.

Answer: D. Five.

48. Identify the primary goal of classification.

- A. Predicting continuous values
- B. Sorting data into groups based on similarity
- C. Reducing the dimensionality of data
- D. Clustering data based on features

Answer: Sorting data into groups based on similarity

49. Which is define Web Mining?

- A. Extracting gold from the internet
- B. Extracting useful information or knowledge from the World Wide Web
- C. Building websites for data collection

D. Creating web-based video games

Answer: Extracting useful information or knowledge from the World Wide Web

50. Identify the approach commonly used to solve a classification problem

- A. Supervised learning
- B. Unsupervised learning
- C. Reinforcement learning
- D. Semi-supervised learning

Answer: Supervised learning

51. Identify how are classifiers typically evaluated

- A. By accuracy only
- B. By precision only
- C. By a combination of metrics such as accuracy, precision, recall, and F1-score
- D. By computational complexity

Answer: By a combination of metrics such as accuracy, precision, recall, and F1-score

52. Identify the classification technique constructs a decision tree based on the attributes of the dataset

- A. K-Nearest Neighbor
- B. Support Vector Machine
- C. Decision Trees
- D. Naive-Bayes Classifier

Answer: Decision Trees

53. Which of the following is not a type of Web Mining in the given list?

- A. Web Content Mining
- B. Web Structure Mining
- C. Web Usage Mining
- D. Web Searching

Answer: Web Searching

54. Determine what is Web Content Mining concerned with?

- A. Analyzing the layout of a webpage
- B. Analyzing the structure of a webpage
- C. Extracting useful information from web documents
- D. Analyzing user behavior on the web

Answer: Extracting useful information from web documents

55. Web Structure Mining match with:

- A. Mining the structure of the internet
- B. Mining the structure of web documents
- C. Mining the structure of web usage patterns
- D. Mining the structure of web servers

Answer: Mining the structure of web documents

56. Determine which of the following is an example of Web Usage Mining?

- A. Analyzing the structure of a webpage
- B. Examining the content of a webpage
- C. Analyzing user access patterns on a website
- D. Extracting information from a database

Answer: Analyzing user access patterns on a website

57. Which of the following is a challenge in Web Mining in the given list?

- A. Lack of data on the web
- B. High-quality data available in abundance
- C. Heterogeneity of web data
- D. Uniformity of web data

Answer: Heterogeneity of web data

58. Select what is the primary goal of Web Content Mining?

- A. Understanding user behavior
- B. Analyzing web usage patterns
- C. Extracting useful information from web documents
- D. Identifying popular websites

Answer: Extracting useful information from web documents

59. Select which algorithm is used for association rule mining in Web Usage Mining?

- A. K-means
- B. Apriori
- C. Decision Trees
- D. Support Vector Machines

Answer: Apriori

60. Select what does TF-IDF stand for in the context of Web Content Mining?

- A. Term Frequency-Inverse Document Frequency
- B. Total Frequency-Inverse Document Factor
- C. Textual Frequency-Inverse Document Format
- D. Term Factor-Inverse Document Formula

Answer: Term Frequency-Inverse Document Frequency

61. Determine which of the following is not a step in the Web Mining process?

- A. Preprocessing
- B. Pattern Discovery
- C. Pattern Analysis
- D. Pattern Recognition

Answer: Pattern Recognition

62. Select which of the following is an example of a web mining tool?

- A. Google Chrome
- B. Microsoft Word
- C. Web Miner
- D. Adobe Photoshop

Answer: Web Miner

63. Which type of Web Mining apply on analyzing user sessions and navigation paths?

- A. Web Content Mining
- B. Web Structure Mining
- C. Web Usage Mining
- D. Web Crawling

Answer: Web Usage Mining

64. Select what is the objective of Web Structure Mining?

- A. To analyze the layout of a webpage
- B. To analyze the content of a webpage
- C. To analyze the hyperlink structure between web pages
- D. To analyze the user behavior on the web

Answer: To analyze the hyperlink structure between web pages

65. Examine which of the following is not a typical application area of Web Mining?

- A. E-commerce
- B. Healthcare
- C. Social Media Analysis
- D. Astrophysics

Answer: Astrophysics

66. Recall what does OLAP stand for in the context of Web Usage Mining?

- A. Online Analytical Processing
- B. Online Learning and Prediction

- C. Offline Analytical Processing
- D. Open Learning and Practice

Answer: Online Analytical Processing

67. Choose what is the main purpose of web crawling in Web Mining?

- A. Extracting information from web documents
- B. Analyzing the structure of web documents
- C. Indexing web pages for search engines
- D. Analyzing user behavior on the web

Answer: Indexing web pages for search engines

68. Select which data mining technique is often used in Web Structure Mining to discover patterns in hyperlink structures?

- A. Decision Trees
- B. Association Rule Mining
- C. Clustering
- D. Sequential Pattern Mining

Answer: Association Rule Mining

69. Determine which aspect of Web Mining involves analyzing user interactions with web pages?

- A. Web Content Mining
- B. Web Structure Mining
- C. Web Usage Mining
- D. Web Crawling

Answer: Web Usage Mining

70. State what is text mining?

- A. Extracting valuable metals from text documents
- B. Extracting useful information or knowledge from text documents
- C. Typing documents for data analysis
- D. Deleting text from documents

Answer: Extracting useful information or knowledge from text documents

71. Select which of the following is not a primary task in text mining?

- A. Information retrieval
- B. Text classification
- C. Sentiment analysis
- D. Image recognition

Answer: Image recognition

72. Determine what is the process of converting unstructured text data into structured or semi-structured data called?

- A. Text preprocessing
- B. Text normalization
- C. Text transformation
- D. Text encoding

Answer: Text preprocessing

73. Select which of the following is not a typical text mining technique?

- A. Clustering
- B. Regression analysis
- C. Topic modeling
- D. Named entity recognition

Answer: Regression analysis

74. Determine which technique is used to extract the main topics or themes from a collection of text documents?

- A. Clustering
- B. Sentiment analysis
- C. Topic modeling
- D. Named entity recognition

Answer: Topic modeling

75. From given list, which of the following is not a common text mining application?

- A. Email spam filtering
- B. Language translation
- C. News categorization
- D. Social media analysis

Answer: Language translation

76. Choose which statistical method is commonly used in text classification?

- A. Linear regression
- B. Principal component analysis
- C. Naive Bayes
- D. K-nearest neighbors

Answer: Naive Bayes

77. Define stemming in text mining.

- A. Identifying the root form of words
- B. Analyzing the sentiment of text

- C. Identifying named entities
- D. Classifying text documents

Answer: Identifying the root form of words

78. State what is the purpose of text summarization in text mining?

- A. Converting unstructured text into structured data
- B. Extracting key information from text documents
- C. Classifying text documents into predefined categories
- D. Identifying and analyzing patterns in text data

Answer: Extracting key information from text documents

79. Tell which algorithm is commonly used for document clustering in text mining?

- A. K-means
- B. Decision Trees
- C. Support Vector Machines
- D. Naive Bayes

Answer: K-means

80. Identify the objective of clustering algorithms.

- A. Grouping similar data points together
- B. Predicting future data points
- C. Identifying outliers
- D. Dimensionality reduction

Answer: Grouping similar data points together

81. Identify a commonly used algorithm for partitioning data into distinct groups.

- A. K-Means
- B. Hierarchical Clustering
- C. DBSCAN
- D. SOM (Self-Organizing Maps)

Answer: K-Means

82. Select a technique used for dimensionality reduction in clustering.

- A. PCA (Principal Component Analysis)
- B. Decision Trees
- C. Logistic Regression
- D. Random Forest

Answer: PCA (Principal Component Analysis)

83. Select the key concept behind the K-Means clustering algorithm.

- A. Iteratively minimizing the sum of squared distances
- B. Maximizing silhouette score
- C. Performing spectral decomposition
- D. Optimizing entropy

Answer: Iteratively minimizing the sum of squared distances

84. Select a primary challenge in hierarchical clustering.

- A. Determining the optimal number of clusters
- B. Handling noise and outliers
- C. Managing computational complexity
- D. Dealing with non-linear data

Answer: Determining the optimal number of clusters

85. Select a method commonly used for evaluating clustering algorithms.

- A. Silhouette score
- B. Accuracy
- C. F1-score
- D. Mean Squared Error

Answer: Silhouette score

86. Identify a technique used for outlier detection.

- A. DBSCAN
- B. PCA
- C. SVM (Support Vector Machine)
- D. KNN (K-Nearest Neighbors)

Answer: DBSCAN

87. Give an example of a divisive method in hierarchical clustering.

- A. DIANA (Divisive Analysis)
- B. Single Linkage
- C. Ward's Method
- D. Complete Linkage

Answer: DIANA (Divisive Analysis)

88. Select the strengths of hierarchical clustering.

- A. Ability to visualize clusters hierarchically
- B. Efficiency with large datasets
- C. Robustness to outliers
- D. Scalability

Answer: Ability to visualize clusters hierarchically

89. Classify an issue in K-Means clustering.

- A. Sensitivity to initial centroid placement
- B. Robustness to outliers
- C. Handling non-linear data
- D. Handling large datasets

Answer: Sensitivity to initial centroid placement

90. Select the role of distance metrics in clustering algorithms.

- A. Measure dissimilarity between data points
- B. Determine optimal number of clusters
- C. Optimize cluster centroids
- D. Assess clustering quality

Answer: Measure dissimilarity between data points

91. Identify a common drawback of K-Means clustering.

- A. Sensitivity to initial centroid placement
- B. Robustness to outliers
- C. Efficient for large datasets
- D. Handles clusters of varying sizes

Answer: Sensitivity to initial centroid placement

92. Identify the significance of outlier detection in clustering.

- A. Ensure the quality of clusters
- B. Improve computational efficiency
- C. Enhance interpretability
- D. Optimize clustering parameters

Answer: Ensure the quality of clusters

93. Select a method used for agglomerative hierarchical clustering.

- A. Single Linkage
- B. K-Means
- C. DBSCAN
- D. PCA

Answer: Single Linkage

94. Identify a technique for dimensionality reduction.

- A. PCA (Principal Component Analysis)
- B. Decision Trees

- C. K-Means
- D. Support Vector Machines

Answer: PCA (Principal Component Analysis)

95. Classify a method used for evaluating the performance of clustering algorithms.

- A. Silhouette Score
- B. Accuracy
- C. F1-Score
- D. Mean Squared Error

Answer: Silhouette Score

96. Select a key issue in hierarchical clustering.

- A. Determining optimal number of clusters
- B. Handling noisy data
- C. Managing computational complexity
- D. Dealing with non-linear data

Answer: Determining optimal number of clusters

97. Identify a strength of K-Means clustering.

- A. Efficiency for large datasets
- B. Robustness to outliers
- C. Handling clusters of varying sizes
- D. Sensitive to initial centroid placement

Answer: Efficiency for large datasets

98. Select a technique commonly used for dimensionality reduction.

- A. PCA (Principal Component Analysis)
- B. Logistic Regression
- C. Decision Trees
- D. K-Nearest Neighbors

Answer: PCA (Principal Component Analysis)

99. Give an example of a divisive hierarchical clustering method.

- A. DIANA (Divisive Analysis)
- B. Ward's Method
- C. Average Linkage
- D. Complete Linkage

Answer: DIANA (Divisive Analysis)

100. Identify a challenge in K-Means clustering.

- A. Sensitivity to initial centroid placement
- B. Handling outliers
- C. Handling non-linear data
- D. Determining optimal number of clusters

Answer: Sensitivity to initial centroid placement

101. Select the process of outlier detection in clustering.

- A. Identify data points deviating from the majority
- B. Group similar data points together
- C. Optimize cluster centroids
- D. Assess clustering quality

Answer: Identify data points deviating from the majority

102. Identify the role of distance metrics in hierarchical clustering.

- A. Measure dissimilarity between clusters
- B. Determine cluster centroids
- C. Optimize number of clusters
- D. Assess clustering quality

Answer: Measure dissimilarity between clusters

103. Classify a technique used for dimensionality reduction in clustering.

- A. PCA (Principal Component Analysis)
- B. Decision Trees
- C. Support Vector Machines
- D. Random Forest

Answer: PCA (Principal Component Analysis)

104. Identify a commonly used method for evaluating clustering algorithms.

- A. Silhouette Score
- B. Accuracy
- C. F1-Score
- D. Mean Squared Error

Answer: Silhouette Score

105. Select a method used for outlier detection.

- A. DBSCAN
- B. K-Means
- C. Agglomerative Hierarchical Clustering
- D. Spectral Clustering

Answer: DBSCAN

106. Identify a clustering algorithm sensitive to the initial placement of centroids.

- A. PAM Algorithm
- B. Hierarchical Clustering
- C. DBSCAN Algorithm
- D. K-Means Algorithm

Answer: K-Means Algorithm

107. Identify a technique used for reducing the dimensionality of data in clustering.

- A. K-Means Algorithm
- B. Hierarchical Clustering
- C. PCA Algorithm
- D. DBSCAN Algorithm

Answer: PCA Algorithm

108. Identify a key issue in hierarchical clustering related to computational complexity.

- A. Difficulty in handling large datasets
- B. Sensitivity to outliers
- C. Inability to handle non-linear data
- D. Inefficiency in memory usage

Answer: Difficulty in handling large datasets

109. Give an example of an additional issue in K-Means clustering apart from sensitivity to initial centroids.

- A. Convergence to local optima
- B. Robustness to noise
- C. Ability to handle non-linear data
- D. Efficiency in large datasets

Answer: Convergence to local optima

110. Select a method used to determine the optimal number of clusters in K-Means clustering.

- A. Elbow Method
- B. Silhouette Analysis
- C. Gap Statistics
- D. All of the these

Answer: All of the these

111. Classify the technique used to handle outliers before clustering.

- A. Outlier Detection

- B. Dimension Reduction
- C. Partitioning
- D. Evaluation of Clustering Algorithms

Answer: Outlier Detection

112. Identify the algorithm commonly used for handling noise and outliers in clustering.

- A. DBSCAN Algorithm
- B. K-Means Algorithm
- C. PAM Algorithm
- D. Hierarchical Clustering

Answer: DBSCAN Algorithm

113. Select a method used to evaluate the compactness within clusters and separation between clusters.

- A. Davies–Bouldin Index
- B. Principal Component Analysis (PCA) Algorithm
- C. K-Means Algorithm
- D. Partitioning

Answer: Davies–Bouldin Index

114. Identify the primary goal of classification. A) Predicting continuous values B) Sorting data into groups based on similarity

- A. Predicting continuous values
- B. Sorting data into groups based on similarity
- C. Reducing the dimensionality of data
- D. Clustering data based on features

Answer: Sorting data into groups based on similarity

115. Identify the algorithm , commonly used for decision tree induction.

- A. K-Means
- B. C4.5
- C. DBSCAN
- D. Random Forest

Answer: C4.5

116. Identify ,What does a Bayes Classifier rely on.

- A. Decision boundaries
- B. Feature independence assumption
- C. Decision trees
- D. Majority voting

Answer: Feature independence assumption

117. Identify ,the classifier assumes independence between features.

- A. Bayes Classifier
- B. Naive-Bayes Classifier
- C. Decision Tree Classifier
- D. K-Nearest Neighbor Classifier

Answer: Naive-Bayes Classifier

118. Choose,the another term for Bayesian Belief Networks.

- A. Markov Networks
- B. Hidden Markov Models
- C. Probabilistic Graphical Models
- D. Conditional Random Fields

Answer: Probabilistic Graphical Models

119. Identify , the primary characteristic of K-Nearest Neighbor classification.

- A. It constructs a decision tree
- B. It requires labeled training data
- C. It uses Bayes\' theorem
- D. It makes predictions based on the majority class of its k-nearest neighbors

Answer: It makes predictions based on the majority class of its k-nearest neighbors

120. Identify the purpose of using kernel functions in Support Vector Machines.

- A. To transform data into a higher-dimensional space
- B. To reduce the dimensionality of the data
- C. To calculate distances between data points
- D. To normalize the data

Answer: To transform data into a higher-dimensional space

121. Identify the following is NOT a classification technique.

- A.) Linear Regression
- B. Logistic Regression
- C. Decision Trees
- D. Support Vector Machines

Answer:) Linear Regression

122. Identify ,In a confusion matrix, what does the true negative (TN) represent.

- A. Predicted negative and actually negative instances
- B. Predicted positive and actually positive instances

- C. Predicted negative but actually positive instances
- D. Predicted positive but actually negative instances

Answer: Predicted negative and actually negative instances

123. Identify ,the evaluation metric penalizes false negatives more than false positives.

- A. Precision
- B. Recall
- C. F1-score
- D. Accuracy

Answer: Recall

124. Identify ,the ensemble learning.

- A. A technique to reduce overfitting by combining multiple models
- B. A method to select the best features for classification
- C. An approach to dimensionality reduction
- D. A way to balance the dataset

Answer: An approach to dimensionality reduction

125. Identify the cross-validation used for in classification

- A. To split the dataset into training and testing sets
- B. To tune hyperparameters of the classifier
- C. To evaluate the performance of the classifier
- D. To preprocess the data before classification

Answer: To evaluate the performance of the classifier

126. Identify the term \"overfitting\" refer to in classification.

- A. The model performs well on unseen data
- B. The model performs poorly on the training data
- C. The model captures noise in the training data
- D. The model is undertrained

Answer: The model captures noise in the training data

127. Identify ,the technique is used to handle imbalanced datasets in classification.

- A. Random sampling
- B. SMOTE (Synthetic Minority Over-sampling Technique)
- C. Dimensionality reduction
- D. Feature scaling

Answer: SMOTE (Synthetic Minority Over-sampling Technique)

128. Identify ,the purpose of feature scaling in classification

- A. To remove irrelevant features
- B. To convert categorical features into numerical ones
- C. To normalize the range of features
- D. To increase the dimensionality of the dataset

Answer: To normalize the range of features

129. Choose the algorithm suitable for handling large-scale classification tasks

- A. K-Nearest Neighbor
- B. Decision Trees
- C. Support Vector Machines
- D. Naive-Bayes Classifier

Answer: Support Vector Machines

130. Choose the main disadvantage of the Naive-Bayes Classifier.

- A. it assumes independence between features
- B. It is sensitive to outliers
- C. It requires a large amount of training data
- D. It cannot handle categorical variables

Answer: it assumes independence between features

131. Choose the main advantage of decision trees.

- A. They are robust to noisy data
- B. They can handle large datasets efficiently
- C. They require little data preprocessing
- D. They are not affected by the curse of dimensionality

Answer: They require little data preprocessing

132. Choose the method used to handle missing data in classification.

- A. Removing instances with missing data
- B. Replacing missing values with the mean or median
- C. Ignoring missing values during training
- D. Using regression to predict missing values

Answer: Replacing missing values with the mean or median

133. Choose the primary limitation of the k-nearest neighbor algorithm.

- A. It requires a large amount of memory
- B. It is sensitive to the choice of distance metric
- C. It cannot handle categorical variables
- D. It is computationally expensive during training

Answer: It is sensitive to the choice of distance metric

134. Choose the kernel function ,commonly used in Support Vector Machines for non-linear classification tasks.

- A. Linear kernel
- B. Polynomial kernel
- C. Sigmoid kernel
- D. Radial Basis Function (RBF) kernel

Answer: Radial Basis Function (RBF) kernel

135. Choose the primary advantage of ensemble methods such as Random Forest.

- A. They are simple and easy to interpret
- B. They are resistant to overfitting
- C. They do not require hyperparameter tuning
- D. They are computationally efficient

Answer: They are resistant to overfitting

136. Choose the technique is used to visualize the decision boundaries of classifiers

- A. Scatter plot
- B. Histogram
- C. ROC curve
- D. Contour plot

Answer: Contour plot

137. Choose the main purpose of regularization in classification algorithms.

- A. To increase the complexity of the model
- B. To reduce the complexity of the model
- C. To eliminate bias in the model
- D. To increase the variance of the model

Answer: To reduce the complexity of the model

138. Choose the method ,used to avoid overfitting in decision trees.

- A. Pruning
- B. Feature selection
- C. Regularization
- D. Ensemble learning

Answer: Pruning

139. Choose , the term \"hyperplane\" refer to in Support Vector Machines.

- A. A line that separates two classes in a 2D space

- B. A plane that separates two classes in a 3D space
- C. A decision boundary in a higher-dimensional space
- D. A curve that separates two classes in a feature space

Answer: A decision boundary in a higher-dimensional space

140. Choose the evaluation metric combines precision and recall into a single value.

- A. Accuracy
- B. F1-score
- C. Area Under the ROC Curve (AUC)
- D. Confusion Matrix

Answer: F1-score

141. Select which of the following steps is NOT part of the data preprocessing stage in KDP?

- A. Data Integration
- B. Data Reduction
- C. Data Mining
- D. Discretization

Answer: Data Mining

142. Select which technique is used for reducing the complexity of data while preserving its integrity in data reduction?

- A. Principal Component Analysis (PCA)
- B. Linear Regression
- C. Random Forest
- D. K-Means Clustering

Answer: Principal Component Analysis (PCA)

143. Select, which one is primarily concerned with Concept hierarchy generation in data mining

- A. Reducing data dimensionality
- B. Creating a hierarchical structure for categorical data
- C. Applying machine learning algorithms
- D. Cleaning noisy data

Answer: Creating a hierarchical structure for categorical data

144. Select, what is the main goal of data discretization?

- A. To convert continuous data into categorical data
- B. To remove outliers from the dataset
- C. To merge similar data points
- D. To standardize the data distribution

Answer: To convert continuous data into categorical data

145. Select the correct option for " Basics of data mining involve understanding:"

- A. Advanced statistical methods
- B. The structure and characteristics of data
- C. Deep learning algorithms
- D. Quantum computing principles

Answer: The structure and characteristics of data

146. Select ,which data mining technique is suitable for finding hidden patterns in large datasets?

- A. Regression analysis
- B. Association rule mining
- C. Decision trees
- D. K-nearest neighbors

Answer: Association rule mining

147. Select, In the KDP process, which stage involves extracting patterns that have meaningful interpretation?

- A. Data Integration
- B. Data Mining
- C. Data Reduction
- D. Concept Hierarchy Generation

Answer: Data Mining

148. Select which of the following is NOT a step in the Knowledge Discovery Process (KDP)?

- A. Data Retrieval
- B. Pattern Evaluation
- C. Pattern Discovery
- D. Data Presentation

Answer: Data Retrieval

149. Select Data integration involves with which one

- A. Transforming data into a suitable format for analysis
- B. Combining data from different sources to provide a unified view
- C. Reducing the amount of data without losing significant information
- D. Creating a hierarchical structure for categorical data

Answer: Combining data from different sources to provide a unified view

150. Select what is the primary purpose of data preprocessing in KDP?

- A. To improve the quality of data
- B. To deploy machine learning models

- C. To visualize data patterns
- D. To generate new data concepts

Answer: To improve the quality of data

151. Choose the step in the KDP process, which one involves removing irrelevant or redundant attributes from the dataset

- A. Data Transformation
- B. Data Reduction
- C. Data Mining
- D. Data Integration

Answer: Data Reduction

152. Choose the technique which is used to transform categorical data into numerical form for data mining purposes?

- A. One-hot encoding
- B. Principal Component Analysis (PCA)
- C. Regression analysis
- D. K-Means clustering

Answer: One-hot encoding

153. Concept hierarchy generation is essential for: Choose the correct option.

- A. Identifying trends in data
- B. Discovering hidden patterns
- C. Reducing the complexity of categorical data
- D. Transforming numerical data into categorical form

Answer: Reducing the complexity of categorical data

154. Choose from the given option which method is employed to predict the class label of a given data instance, In the context of data mining techniques?

- A. Clustering
- B. Association rule mining
- C. Classification
- D. Regression analysis

Answer: Classification

155. Choose from the given option which data mining technique focuses on identifying frequent itemsets in a dataset?

- A. Clustering
- B. Classification
- C. Association rule mining

D. Regression analysis

Answer: Association rule mining

156. Select ,which algorithm is commonly selected for association rule mining in large databases?

A. K-means

B. Decision Trees

C. Apriori

D. SVM

Answer: Apriori

157. Select, what is the primary purpose of selecting association rule mining?

A. Clustering similar data points

B. Predicting continuous values

C. Discovering relationships between variables

D. Classifying data into predefined categories

Answer: Discovering relationships between variables

158. Select from the following,How would you select the support value for a specific itemset from a large transaction dataset?

A. How would you select the support value for a specific itemset from a large transaction dataset?

B. Calculate the ratio of transactions containing the itemset to the total transactions

C. Use a pre-defined threshold for support d) Apply statistical tests to estimate support

D. none

Answer: Calculate the ratio of transactions containing the itemset to the total transactions

159. Select, what is a fundamental step selected in association rule mining?

A. Feature engineering

B. Dimensionality reduction

C. Support counting

D. Model training

Answer: Support counting

160. Select from the following why is the Apriori algorithm selected for mining association rules in large databases?

A. It has a low computational complexity

B. It can handle noisy data effectively

C. It efficiently prunes the search space

D. It doesn't require prior knowledge about the data

Answer: It efficiently prunes the search space

161. Suppose you have a dataset with a large number of transactions and items. Which technique would you Choose to remove infrequent items efficiently? Select from the following.

- A. Principal Component Analysis (PCA)
- B. Support Vector Machines (SVM)
- C. Apriori algorithm
- D. Linear Discriminant Analysis (LDA)

Answer: Apriori algorithm

162. Select from the following,how does the concept of minimum support play a crucial role in association rule mining?

- A. It determines the maximum number of transactions in a dataset
- B. It sets a threshold for the frequency of itemsets to be considered frequent
- C. It defines the maximum number of rules generated by the algorithm
- D. It measures the strength of association between items in a transaction

Answer: It sets a threshold for the frequency of itemsets to be considered frequent

163. Select a strategy to handle sparsity issues in association rule mining when dealing with large datasets

- A. Increase the minimum support threshold
- B. Decrease the minimum support threshold
- C. Perform feature engineering to reduce sparsity
- D. Apply dimensionality reduction techniques

Answer: Perform feature engineering to reduce sparsity

164. Choose from the following what is the primary goal selected of mining association rules?

- A. Clustering similar data points
- B. Predicting continuous values
- C. Identifying relationships between variables
- D. Classifying data into predefined categories

Answer: Identifying relationships between variables

165. Choose from the following,how does the Apriori algorithm handle candidate itemset generation?

- A. It generates all possible itemsets and then prunes based on support
- B. It generates itemsets randomly and evaluates their support
- C. It generates itemsets based on user-defined criteria
- D. It generates itemsets iteratively based on existing frequent itemsets

Answer: It generates itemsets iteratively based on existing frequent itemsets

166. Suppose you're tasked with analyzing market basket data to identify purchasing patterns. Choose from the following, how would you select appropriate metrics to evaluate the discovered association rules?

- A. By considering measures like support, confidence, and lift
- B. By using random sampling techniques
- C. By applying clustering algorithms
- D. By visualizing the data using scatter plots

Answer: By considering measures like support, confidence, and lift

167. Considering a scenario where you need to find association rules in multi-level datasets, Choose from the following, which approach would you select and why?

- A. Apriori algorithm, for its ability to handle large datasets efficiently
- B. K-means clustering, for its simplicity in implementation
- C. Decision Trees, for their interpretability
- D. Naive Bayes, for its probabilistic approach

Answer: Apriori algorithm, for its ability to handle large datasets efficiently

168. Choose from the following the advantages and disadvantages of selecting single-dimensional Boolean association rules versus multi-level association rules.

- A. Single-dimensional rules are easier to interpret, but multi-level rules provide more detailed insights.
- B. Single-dimensional rules are computationally less expensive, but multi-level rules can capture complex relationships.
- C. Single-dimensional rules are more accurate, but multi-level rules have higher scalability.
- D. Single-dimensional rules are prone to overfitting, but multi-level rules offer better generalization.

Answer: Single-dimensional rules are computationally less expensive, but multi-level rules can capture complex relationships.

169. Choose the impact of selecting different minimum support thresholds on the number and quality of discovered association rules.

- A. Lower support thresholds result in fewer but more significant rules
- B. Higher support thresholds lead to more rules with lower confidence
- C. The choice of support threshold does not affect the number or quality of rules
- D. Adjusting the support threshold has no impact on the rule generation process

Answer: Higher support thresholds lead to more rules with lower confidence

170. Identify the functions of Data Mining.

- A. Association and correctional analysis classification.
- B. Prediction and characterization.

- C. Cluster analysis and Evolution analysis.
- D. All of the these.

Answer: All of the these.

171. Select the essential process in which the intelligent methods are applied to extract data patterns.

- A. Warehousing
- B. Data Mining
- C. Text Mining
- D. Data Selection

Answer: Data Mining

172. Choose the one that can be considered as the correct process of Data Mining.

- A. Infrastructure, Exploration, Analysis, Interpretation, Exploitation.
- B. Exploration, Infrastructure, Analysis, Interpretation, Exploitation.
- C. Exploration, Infrastructure, Interpretation, Analysis, Exploitation.
- D. Exploration, Infrastructure, Analysis, Exploitation, Interpretation.

Answer: Infrastructure, Exploration, Analysis, Interpretation, Exploitation.

173. Select the one that refers to querying the unstructured textual data.

- A. Information access.
- B. Information update.
- C. Information retrieval.
- D. Information manipulation.

Answer: Information retrieval.

174. Identify that refers to the problem of finding abstracted patterns (or structures) in the unlabeled data.

- A. Supervised learning
- B. Unsupervised learning
- C. Hybrid learning
- D. Reinforcement learning

Answer: Unsupervised learning

175. Select the statements that is incorrect about the hierarchal clustering.

- A. The hierarchal type of clustering is also known as the HCA
- B. The choice of an appropriate metric can influence the shape of the cluster
- C. In general, the splits and merges both are determined in a greedy manner
- D. All of these.

Answer: All of these.

176. Select the statement. that is correct about K-means clustering.

- A. The goal of the k-means clustering is to partition (n) observation into (k) clusters.
- B. K-means clustering can be defined as the method of quantization.
- C. The nearest neighbor is the same as the K-means.
- D. All of these.

Answer: The nearest neighbor is the same as the K-means.

177. Identify the clustering technique that needs the merging approach.

- A. Partitioned
- B. Naïve Bayes
- C. Hierarchical
- D. Both A and C

Answer: Hierarchical

178. Identify the self-organizing maps can also be considered as the instance

- A. Supervised learning
- B. Unsupervised learning
- C. Missing data imputation
- D. Both A & C

Answer: Unsupervised learning

179. Select the statement that is true about the classification.

- A. It is a measure of accuracy
- B. It is a subdivision of a set
- C. It is the task of assigning a classification
- D. None of these

Answer: It is a subdivision of a set

180. Select OLAP operation that allows users to view data at a more details level.

- A. Roll up
- B. Slice
- C. Drill-down
- D. Dice

Answer: Drill-down

181. Select the component of a data warehouse that provides a summarized, department-focused view.

- A. OLAP
- B. Data Lake

C. Data Mart

D. MetaData

Answer: Data Mart

182. Identify the process that extracts, converts, and loads data into a data warehouse.

A. OLAP

B. ETL

C. NoSQL

D. DDL

Answer: ETL

183. Choose the main purpose of a data warehouse.

A. manage real time messaging.

B. Run transactional operation

C. Analyze historical data for decision making.

D. Backup operational system.

Answer: Analyze historical data for decision making.

184. Data transformation is mainly used to

A. Delete data

B. Encrypt data

C. Convert data into suitable format

D. Archive data

Answer: Convert data into suitable format

185. Select component that stores information about data definitions and mappings.

A. OLAP server

B. Metadata repository

C. Data mart

D. Source system

Answer: Metadata repository

186. Select the schema that offers the highest query performance.

A. Snowflake

B. Galaxy

C. Star

D. Network

Answer: Star

187. Incremental extraction refers to

- A. Extracting all data.
- B. Extracting summarized data.
- C. Extracting only changed data.
- D. Extracting transformed data.

Answer: Extracting only changed data.

188. The conceptual design of a data warehouse mainly focuses on

- A. Indexing strategies
- B. Business requirements and high-level data view
- C. Physical storage structures
- D. Query optimization

Answer: Business requirements and high-level data view

189. Logical design of a data warehouse primarily defines

- A. Disk allocation
- B. File organization
- C. Tables, attributes, and relationships
- D. Backup strategies

Answer: Tables, attributes, and relationships

190. Snowflake schema differs from star schema because it

- A. Uses more fact tables
- B. Normalizes dimension tables
- C. Eliminates dimensions
- D. Improves query speed

Answer: Normalizes dimension tables

191. Physical design of a data warehouse deals with

- A. Business rules
- B. Data modeling
- C. Storage, indexing, and partitioning
- D. Requirement analysis

Answer: Storage, indexing, and partitioning

192. Fixing a single dimension value to view a subset of data is called

- A. Dice
- B. Slice
- C. Pivot
- D. Roll-up

Answer: Slice

193. Rotating the data cube to change the data view is known as

- A. Drill-down
- B. Roll-up
- C. Pivot
- D. Slice

Answer: Pivot

194. ROLAP primarily uses

- A. Multidimensional databases
- B. Object-oriented databases
- C. Relational databases
- D. NoSQL databases

Answer: Relational databases

195. Support and confidence are measures used in

- A. Classification
- B. Clustering
- C. Association rule mining
- D. Regression

Answer: Association rule mining

196. Data warehouses primarily support data mining by providing

- A. Real-time transaction data
- B. Clean, integrated, historical data
- C. Unstructured data
- D. Temporary data

Answer: Clean, integrated, historical data

197. Recognize the model used to predict future outcomes

- A. Descriptive model
- B. Logical model
- C. Predictive model
- D. Conceptual model

Answer: Predictive model

198. State the measure used in association rule mining.

- A. Accuracy
- B. Confidence

C. Error rate

D. Variance

Answer: Confidence

199. Identify the data mining task used to group similar objects.

A. Classification

B. Regression

C. Clustering

D. Prediction

Answer: Clustering

200. Select the task that assigns class labels to data.

A. Clustering

B. Classification

C. Association

D. Summarization

Answer: Classification

201. Recognize the correct definition of Knowledge Discovery Process (KDP).

A. Process of database creation.

B. Process of extracting useful knowledge from data.

C. Process of data encryption.

D. Process of query optimization.

Answer: Process of extracting useful knowledge from data.

202. Select the first step in the KDD process.

A. Data mining.

B. Data cleaning.

C. Pattern evaluation.

D. Knowledge presentation.

Answer: Data cleaning.

203. Identify the purpose of data preprocessing.

A. Increase data redundancy

B. Improve data quality

C. Encrypt data

D. Store data permanently

Answer: Improve data quality

204. Choose the task that removes noise and inconsistencies from data.

- A. Data integration
- B. Data reduction
- C. Data cleaning
- D. Data discretization

Answer: Data cleaning

205. Recognize data integration as a process that performs

- A. Pattern extraction
- B. Data compression
- C. Combining multiple data sources
- D. Rule generation

Answer: Combining multiple data sources

206. Identify normalization as a technique under

- A. Data reduction
- B. Data transformation
- C. Data discretization
- D. Data mining

Answer: Data transformation

207. Select the goal of data reduction.

- A. Increase data size
- B. Improve visualization
- C. Reduce data volume without losing information
- D. Convert continuous data

Answer: Reduce data volume without losing information

208. Identify PCA as a technique used for

- A. Data integration
- B. Dimensionality reduction
- C. Association rule mining
- D. Data cleaning

Answer: Dimensionality reduction

209. Recognize discretization as a process that converts

- A. Categorical data to numeric
- B. Numeric data to continuous
- C. Continuous data to discrete intervals
- D. Discrete data to continuous

Answer: Continuous data to discrete intervals

210. Select an example of a concept hierarchy.

- A. Age $\rightarrow$ Salary $\rightarrow$ Experience
- B. City $\rightarrow$ State $\rightarrow$ Country
- C. Name $\rightarrow$ Address $\rightarrow$ Phone
- D. ID $\rightarrow$ Roll $\rightarrow$ Rank

Answer: City $\rightarrow$ State $\rightarrow$ Country

211. Identify classification as a data mining technique used to

- A. Group similar objects
- B. Assign predefined class labels
- C. Reduce dimensions
- D. Detect noise

Answer: Assign predefined class labels

212. Select the key property used in the Apriori algorithm.

- A. Downward closure property
- B. Transitivity rule
- C. Bayesian rule
- D. Classification rule

Answer: Downward closure property

213. Recognize clustering as a technique that performs

- A. Label assignment
- B. Pattern ranking
- C. Similarity-based grouping
- D. Rule generation

Answer: Similarity-based grouping

214. Identify the general form of an association rule

- A. $X + Y$
- B. $X \rightarrow Y$
- C. $X = Y$
- D. $X - Y$

Answer: $X \rightarrow Y$

215. Select the measure that indicates frequency of an itemset.

- A. Confidence
- B. Lift

- C. Support
- D. Conviction

Answer: Support

216. Recognize single-dimensional Boolean association rules as rules that involve

- A. Multiple predicates
- B. One predicate with Boolean values
- C. Continuous attributes
- D. Temporal data

Answer: One predicate with Boolean values

217. Identify the main drawback of the Apriori algorithm.

- A. Inaccurate results
- B. Multiple database scans
- C. Lack of pruning
- D. Poor rule generation

Answer: Multiple database scans

218. Recognize fraud detection as an application of data mining in

- A. Education
- B. Healthcare
- C. Finance
- D. Agriculture

Answer: Finance

219. A data cube in data mining organizes data in

- A. Sequential files
- B. Two-dimensional tables
- C. Multidimensional space
- D. Linear arrays

Answer: Multidimensional space

220. The numerical values stored in a data cube are called

- A. Dimensions
- B. Attributes
- C. Measures
- D. Keys

Answer: Measures

221. Select the OLAP operation that summarizes data along a dimension.

- A. Drill-down
- B. Slice
- C. Roll-up
- D. Pivot

Answer: Roll-up

222. Select the challenge associated with high-dimensional data.

- A. Reduced storage cost
- B. Curse of dimensionality
- C. Improved accuracy
- D. Simplified computation

Answer: Curse of dimensionality

223. Recognize the application of association rule mining

- A. Text compression
- B. Market basket analysis
- C. Image rendering
- D. Network routing

Answer: Market basket analysis

224. Recognize the role of concept hierarchy in data mining.

- A. Noise removal
- B. Data compression
- C. Multi-level pattern discovery
- D. Attribute deletion

Answer: Multi-level pattern discovery

225. Select the measure that indicates rule reliability in association mining.

- A. Support
- B. Confidence
- C. Coverage
- D. Conviction

Answer: Confidence

226. Recognize the purpose of data reduction.

- A. Increase data size
- B. Improve data visualization only
- C. Reduce data volume with minimal information loss
- D. Eliminate all attributes

Answer: Reduce data volume with minimal information loss

227. Recognize the process that combines data from multiple sources.

- A. Data discretization
- B. Data integration
- C. Data reduction
- D. Data mining

Answer: Data integration

Section 2: SAQ (Short Answer Questions)

1. Select one data pre-processing technique and explain how it helps in improving data quality for data mining purposes.
2. Describe how data integration contributes to data mining by combining data from multiple sources. Provide an example to illustrate your explanation.
3. Describe the process of data reduction and its importance in simplifying datasets for efficient data mining.
4. Discuss two different methods of data discretization commonly used in data mining.
5. Discuss the significance of data transformation in data mining.
6. List down what are the main types of web mining.
7. Describe that how web content mining differ from web structure mining.
8. Define text mining.
9. Define TF-IDF and how is it used in text mining?
10. Write down the challenges which are associated with web mining.
11. Explain how does web usage mining play in improving online user experience.
12. Determine how is text mining applied in real-world scenarios.
13. Explain the significance of discretization in data mining.
14. Write about the basics of data mining and its applications.
15. Explain the concept of association rule mining in data mining.
16. Explain the Knowledge Discovery Process (KDP) in data mining.
17. Identify the key steps involved in concept hierarchy generation.
18. Describe the role of regression analysis in data mining.

19. Define Data Warehouses and their types in detail.
20. What are the steps for the design of a data warehouse?
21. List the Data Warehouse Characteristics in detail
22. Define which one is faster, Multidimensional OLAP or Relational OLAP.
23. Name some popular data warehousing tools and technologies.
24. Explain the concept of ETL in data warehousing.
25. Describe the role of a data mart in a data warehouse environment.
26. Discover the various techniques of data summarization in a data warehouse with a neat diagram.
27. Analyse and design a data warehouse schema for an extensive, complex data set.
28. Differentiate the difference between the dimension table and the fact table.
29. Describe the problem definition in clustering.
30. Explain an overview of clustering techniques.
31. Differentiate between partitioning and hierarchical clustering.
32. Explain the evaluation process of clustering algorithms.
33. Describe the K-Means algorithm.
34. Compare K-Means with PAM algorithm.
35. Analyze the additional issues in K-Means.
36. Describe the basic Agglomerative Hierarchical Clustering Algorithm.
37. Differentiate between agglomerative and divisive methods in hierarchical clustering.
38. Explain key issues in hierarchical clustering.
39. Compare the strengths and weaknesses of hierarchical clustering.
40. Describe the concept of outlier detection.
41. Explain the PCA algorithm for dimension reduction.
42. Analyze the role of PCA in clustering.
43. Describe specific techniques used in hierarchical clustering.
44. Differentiate between outlier detection and clustering.

45. Explain the importance of selecting an appropriate number of clusters in clustering.
46. Compare the PCA algorithm with other dimension reduction techniques.
47. Describe the impact of noisy data on clustering algorithms.
48. Differentiate between hierarchical clustering and non-hierarchical clustering.
49. Explain how hierarchical clustering can handle variable cluster shapes and densities.
50. Describe how outlier detection can be integrated with clustering.
51. Compare agglomerative and divisive methods in hierarchical clustering.
52. Explain how PCA addresses the curse of dimensionality.
53. Analyze the computational complexity of clustering algorithms.
54. Describe the process of selecting the optimal number of clusters in K-Means.
55. Explain the impact of data normalization on clustering results.
56. Describe the role of distance metrics in clustering algorithms.
57. Differentiate between PCA and t-SNE for dimensionality reduction.
58. Explain the impact of outliers on hierarchical clustering.
59. Discuss main objective of classification.
60. Explain the performance of classifiers typically evaluated.
61. Explain a decision tree in classification.
62. Discuss the algorithm commonly used for decision tree induction.
63. Explain the assumption that the Naive-Bayes Classifier makes about the features
64. Explain a Bayesian Belief Network (BBN).
65. Explain the K-Nearest Neighbor (KNN) classification algorithm.
66. Explain the purpose of kernel functions in Support Vector Machines (SVM).
67. Justify the performance of classifiers typically evaluated.
68. Express the curse of dimensionality, and how does it affect classification.
69. Express regularization in classification algorithms.
70. Express a commonly used approach for solving a classification problem, and how is it typically evaluated.

71. Illustrate the concept of a Decision Tree using an experiment on student pass/fail classification. Construct a decision tree based on given conditions and calculate the entropy and information gain for splitting the dataset.
72. Illustrate the K-NN Classification with the given data
73. Explain the fundamental concepts that form the foundation of data mining.
74. Explain the Knowledge Discovery in Databases process and its significance in data mining.
75. Explain Discretization and Concept Hierarchy Generation.
76. Explain Data Reduction.
77. Explain Data Integration and Data Transformation.
78. Consider the following transaction database consisting of four transactions. Transaction with ID 100 contains items 1, 3, and 4. Transaction with ID 200 contains items 2, 3, and 5. Transaction with ID 300 contains items 1, 2, 3, and 5. Transaction with ID 400 contains items 2 and 5. The minimum support threshold is 50% and the minimum confidence threshold is 70%. Apply the Apriori algorithm to identify all frequent itemsets and generate the strong association rules that satisfy the given thresholds.
79. Cluster text documents automatically.
80. Generate association rules and evaluate confidence and lift.
81. Create a classifier where false negatives cost more than false positives and evaluate its impact.
82. Evaluate Euclidean vs Manhattan distance in KNN.
83. A collection contains 6 documents. The term “data” appears 3 times in Document D1, which contains 12 total words. The term appears in 4 documents in the collection. Calculate the TF-IDF value of the term “data” in D1.
84. Two documents are represented by the vectors: $D1 = (1,2,3)$ $D2 = (2,1,2)$ Calculate the Cosine Similarity between the documents.
85. suppose we have two cluster centers: $C_1 = (2, 3)$ $C_2 = (6, 7)$ A document vector is: $P = (4, 5)$ Determine which cluster P belongs to using Euclidean distance.
86. Suppose there are 5 documents. The term “mining” appears 3 times in Document D1 which contains 10 words. The word “mining” appears in 4 documents.

Section 3: LAQ (Long Answer Questions)

1. Define web mining, and how does it differ from traditional data mining.

2. Describe the ethical considerations in web mining and how they can be addressed.
3. Describe how can web mining contribute to knowledge discovery and decision-making in various domains.
4. List down the challenges associated with text mining, and how can they be addressed.
5. Explain the three main types of web mining and provide examples of their applications.
6. Illustrate the main objectives of text mining.
7. Write down the key steps involved in the text mining process.
8. Explain how does text mining contribute to business intelligence and decision-making.
9. Explain the concept of partitioning in clustering algorithms.
10. Describe the K-Means algorithm and its basic steps.
11. Explain the concept of hierarchical clustering and its two main approaches.
12. Explain the PCA algorithm and its objective.
13. Compare and contrast agglomerative and divisive methods in hierarchical clustering.
14. Discuss why dimension reduction is important in data mining.
15. Discuss the main steps involved in PCA.
16. Evaluate the concept of divisive hierarchical clustering and its challenges.
17. Illustrate the trade-offs involved in using PCA for dimension reduction.
18. Estimate the Principal Component Analysis (PCA) algorithm used for in-dimension reduction.
19. Analyze some common methods for identifying outliers in a dataset.
20. Analyze how can outlier detection be performed using clustering techniques.
21. Explain some common applications of dimension reduction techniques in data mining.
22. Discuss role of concept hierarchy generation in a data organization and the steps of concept hierarchy generation.
23. Discuss the steps involved in concept hierarchy generation and its role in data organization.
24. Explain the challenges associated with data mining in modern applications.
25. Identify the role of data mining techniques in fraud detection and prevention.

26. Explain the application of data mining in marketing and customer relationship management (CRM).
27. Explain the significance of association rule mining in data mining
28. Identify the characteristics of single-dimensional boolean association rules.
29. Write about the multi-level association rule mining technique and its applications.
30. Explain the Apriori algorithm for association rule mining.
31. write about the challenges associated with mining association rules in large databases.
32. Write the applications of association rule mining in real-world scenarios.
33. Describe the steps involved in data mining when viewed as a knowledge discovery process
34. List and describe the five primitives for specifying the data mining and warehousing tasks.?
35. Define data cleaning. Express the different techniques for Understanding in handling the missing values.
36. Discuss which algorithm is an influential algorithm for mining frequent item sets for Boolean association rules. Explain with an example.
37. Explain how to mine the multi-dimensional association rules from relational databases and data warehouses.
38. Explain the APRIORI algorithm with an example.
39. Explain the process of solving a classification problem and discuss the key steps involved.
40. Explain the construction process of a decision tree in classification, including the criteria used for node splitting.
41. Explain the Naive-Bayes Classifier, including its underlying assumptions and how it makes predictions.
42. Explain the Bayesian Belief Networks (BBNs) and discuss their role in classification tasks.
43. Express the K-Nearest Neighbor (KNN) classification algorithm, including its algorithmic steps and characteristics.
44. Express the concept of Support Vector Machines (SVMs) in classification, including how they work and their advantages.
45. Define Data Warehouses and Explain their types in detail.
46. Explain the Evolution of Data Warehouses and identify the Data Analytics to AI and Machine Learning in detail.

47. Explain the common OLAP type in detail.
48. What are some real-world applications of KDD? Explain in detail.
49. Enumerate the process of data warehouse in data mining, and what are its key stages? Explain in detail.
50. Compare and Discuss the Key Concepts of Dataware House components in real-world
51. Differentiate fact and Dimensional table in Datawarehouse with example
52. Justify the role of kernel functions in Support Vector Machines (SVMs) and provide examples of commonly used kernels.
53. Express the process of evaluating classifiers and discuss the various metrics used for evaluation.
54. Express the concept of overfitting in classification models and how it can be addressed.
55. Express the curse of dimensionality in the context of classification and its implications.
56. Express the process of feature selection in classification and discuss its importance.
57. Evaluate PCA and reduce to principal component.
58. Illustrate the concept of a confusion matrix using an experiment on a spam email classification model. Compute accuracy, precision, recall, and F1-score based on the given confusion matrix.
59. Explain an apriori algorithm with an example in the given below: Transaction ID Items Bought T1 Milk, Bread, Butter T2 Bread, Butter T3 Milk, Bread T4 Milk, Butter T5 Bread, Butter
60. Explain Snowflake Schema with advantages and limitations for data warehouse design.
61. Define and list the characteristics of Star Schema in multidimensional databases
62. Analyze and compare Star, Snowflake, and Galaxy schemas for multidimensional databases with respect to structure, performance, and usage scenarios.
63. Explain Drill-Down and Roll-Up operations in OLAP
64. Explain Association Rule Mining and Apriori Algorithm.
65. Explain the Knowledge Discovery Process (KDP) with neat diagram.
66. Problem: Build a Decision Tree classifier for customer churn prediction. Requirements: Handle missing values Apply entropy-based split Perform cost-complexity pruning Compare accuracy before and after pruning

67. Create a hybrid ensemble combining: Naive Bayes KNN Use majority voting.
68. Create clustering model that automatically finds optimal number of clusters using Silhouette Score.
69. Cluster user browsing sessions to identify behavior patterns.